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Technical Note

Automatic body mass index detection using correlation of face visual cues

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Abstract. Body mass index (BMI) is used widely as an indicator in general health. Determination of BMI using non-intrusive measurements are of interest and recent advancements in the availability of digital imaging sensors have paved the way for performing quick and automatic measurements. In this work, we consider automatic computation of BMI using correlation features from face images. We show that using face detection based facial fiducial points analysis provides good BMI prediction. Experimental results on comparing the correlation coefficients of facial ratios along with the colour feature has higher significance in BMI of a person.

Keywords: Body mass index (BMI), facial features, correlation coefficient

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1. Introduction

Recent research indicate that the sedentary lifestyles lead to the decline of general health in humans. The general drop in overall human health requires attention towards lifestyle related diseases and disorders [1]. Even with the increase of education and general wellness of societies, the number of disorders is steadily on the rise. The number of people in the obese category had the highest mortality rate [2].

Recently, the level of obesity had reached an astounding average body mass index (BMI) of 20:9% against the 19:8% level observed in 2000 with BMI > 30 is associated with obesity [3], see Table 1. Note that BMI is computed as the ratio of body mass and square of the body height and it is not a scientific

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Table 1
Standard body mass index (BMI) classification system

BMI	Class
< 18.5	Underweight
18.5–24.9	Normal
25.0–29.9	Overweight
30.0–34.9	Obesity (1st Class)
35.0–39.9	Obesity (2nd Class)
> 40	Extreme obesity (3rd Class)

measure rather a rule of thumb for weight based categorization. Obesity is one of the leading causes for related diseases such as higher rate of diabetes and a higher rate of orthopedic related diseases [4].

BMI and weight related issues are still associated with a stigma and it is difficult for medical professionals to make an accurate diagnosis unless all the information about the patient is known.

Considerable research has been done in the area of visual feature extraction, which has enabled a number of non-intrusive methods of feature extraction [5]. This feature extraction has enabled numerous software solutions such as gender determination from image analytics [6]. The techniques have been used to perform research for BMI detection. There has been a correlation between BMI and body fat index which is an important indicator to various diseases. The BMI has also been shown to vary widely for people from different ethnicities [7]. The organization of this communication is as follows.

In Section 2, a related literature review of works is presented. In Section 3 an overview of the proposed technique and the methodology used is presented. The performance evaluation and advantages of the proposed method is shown in Section 4. The conclusion is presented in Section 5.

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2. Literature review

There have been various studies that show that there is a relation between the human face characteristics and general health [8]. They have also showed that the facial characteristics have been disregarded in the past and are an important indicator for health determination. It is forming a basis to determine the quantifiable metrics that can be determined from a face. In [9] the quantifiable metrics are researched about. These metrics are obtained from psychology studies and are tested. We understand that the facial ratios width-height, perimeter-area and cheek-jaw are in proportion with the BMI. This implies that the correlation coefficient is high for the relation.

Finkelstein et al. [10] studied the relation between obesity related diseases insurance spending. They show that the annual spending for insurance companies for the obesity related conditions are worth billions of dollars. The spending was as high as $72.5 Billions even in 1998. A study that uses the results from [8,9] is shown in [11]. The analysis is made on a feature extraction mechanism. The model uses the MORPH-II dataset[1] and is used to determine the BMI of the person. The model does not account for ethnic groups as this is to be attempted in the future.

In [12,13] it has been shown that there is a direct relation between the melanosome level and the ethnicity of a person. The relation between the ethnicity of a person and the effect on the metabolism has also been debated. It has been shown that colour is a major factor to determine the ethnicity of a person [14]. The ethnicity of a person is a direct indicator and can be discerned from face structures.

[1] https://ebill.uncw.edu/C20231 ustores/web/classic/product detail.jsp?PRODUCTID=8.

3. Technique employed

The primary goal that we are aiming to achieve is to provide an efficient visual cues based method that utilizes automatic computer vision techniques for faces. Facial features have been previously found in psychology studies of human faces that are related to general health. The correlation between these face visual features and the BMI is verified to determine the usefulness in our proposed model. The correlation coefficient is given by,

$$r = \frac{\sum(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{\sum(X_i - \overline{X})^2 \sum(Y_i - \overline{Y})^2}} \tag{1}$$

Additionally, an external feature has been added to determine the ethnicity of a person. Research has shown that the ethnicity of the person has a direct relation with the skin colour. We have also used multiple correlation to determine whether certain features are more pronounced when it comes to the ethnicity. The formula for this is given by,

$$R = \frac{\sqrt{r_{yx_1}^2 + r_{yx_2}^2 - 2r_{yx_1}(r_{yx_2})(r_{x_1x_2})}}{\sqrt{1 - r_{x_1x_2}^2}} \tag{2}$$

Our proposed method to determine the BMI of a person does not require any intrusive measurements and do not require the determination of the body weight and the height. We exploit the relationship between the various important facial feature ratios and the BMI of an individual. The following are the ratios utilized here:

- Cheekbone-jaw width (CJWR)
- Width-upper facial height ratio (WHR)
- Perimeter-area ratio (PAR)
- Eye size (ES)

Algorithm 1 BMI Detection Algorithm

Procedure BMIViaVisualCues
Loop:

1. Face detected using Active Shape Model (ASM).
2. Sixty-eight fiducial points extracted to perform further analysis.
   (a) *CJWR* $\leftarrow \frac{p1p17}{p6p12}$
   (b) *WHR* $\leftarrow \frac{p6p12}{p28p58}$
   (c) *PAR* $\leftarrow \frac{Perimeter[p1p6p12p17]}{Area[p1p6p12p17]}$
   (d) *ES* $\leftarrow \frac{1}{2}[p37p46 - p40p43]$
   (e) *LF/FH* $\leftarrow \frac{p29p9}{Np9}$
   (f) *FW/LFH* $\leftarrow \frac{p1p17}{p29p9}$
   (g) *MEH* $\leftarrow 1/6[(p19p37) + (p20p38) + (p22p40) + (p23p43) + (p25p45) + (p26p46)$
3. *Colour* $\leftarrow {}_1^{68}\sum N(pixel_{val})$
4. Compute correlation coefficient for ratios
5. Compute multiple correlation for skin colour:
   (a) $x \leftarrow$ facial ratio (independent variable)
   (b) $y \leftarrow$ colour of person (independent variable)
   (c) $z \leftarrow$ BMI (dependent variable)
6. goto top.

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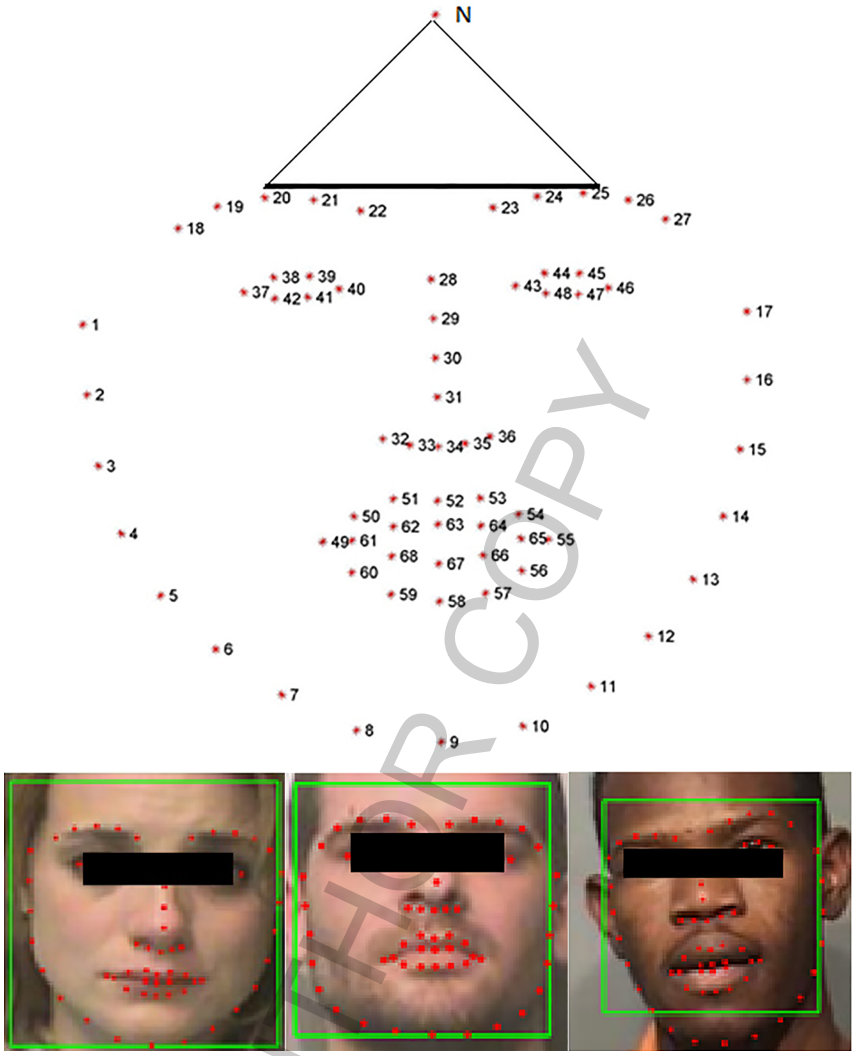

Fig. 1. The 68 facial landmark points, and the facial landmark determination on three different ethnicity faces.

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- Lower face- face height ratio (LF/FH)
- Face width-lower face height ratio (FW/LFH)
- Mean eyebrow height (MEH)

The Fig. 1 depicts the facial landmarks that are used to compute all the ratios, and three examples of detection of the points on different ethnicity subject's faces. As the first step of the detection process requires facial detection and detecting the facial points.

4. Experimental results

The correlation of each of the features has been computed to determine the significance of the model. This process enables us to determine the factors responsible for BMI prediction. The factors determined

Table 2
Comparison with previous based on correlation coefficients. Our method obtained best results overall for using the facial landmark points based ratios in determining the BMI

	Pham et al. [15]	Wen et al. [11]	Our work
PAR	−0.23	−0.07	−0.104
WHR	0.28	0.25	0.37
CJWR	−0.29	−0.2	−0.28
ES	0.26	−0.1	0.067
LFFH	0.1	−0.07	−0.039
FWFLH	0.05	0.15	0.24
MEH	0.16	0.04	0.05

to be of higher significance will be used for further prediction. The results obtained are for CJWR −0.28293, WHR 0.371826, PAR −0.1039, ES 0.066508, LF/FH −0.0389, FW/LFH 0.243652, MEH 0.052856, these values show that there is a correlation between the facial ratios and the BMI of a person. Even though the correlation coefficient is not very strong, it is of significance as shown in the previous works related to facial ratios. The work demonstrated by Pham et al. [15] and Wen and Guo [11] is shown in comparison with our work.

Our findings based on the multiple correlation coefficient with these previous works is presented in Table 2. We have computed the multiple correlation coefficient for each of the ratios and the colour of a person as the independent variables and the BMI as the dependent variable. The results obtained are for CJWR 0.309629, WHR 0.375425, PAR 0.181709, ES 0.176286, LF/FH 0.178765, FW/LFH 0.260871, MEH 0.176634, these values show that the facial ratios when used with the colour of a person bears a higher significance as compared to a single correlation between the ratios and the BMI. Further exploration of different colour spaces on the BMI computation is an interesting ongoing work.

5. Conclusion

We have shown the relation between the various facial features and the BMI. The applications of the proposed model are endless. We have shown how the skin colour of a person plays a role along with the other facial features. This goes to show that significant headway could be made if further studies are conducted in this area. In the future we plan to explore certain other features such as the age of a person or metabolism that could play an important role in determining the BMI of a person.

Conflict of interest

None to report.

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